

Quiz (Fluid Mechanics-BCE23305T)

Total points 30/30 ?

Time: 20 Minutes

Marks: 30

****All questions having equal marks**

Total Questions:15

Semester: Third

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Email *

✓ 4. The property by virtue of which liquid opposes relative motion between its different layers is called 2/2

- ☐ Surface Tension
- ☐ Cohesion
- ☒ Viscosity
- ☐ Capillarity

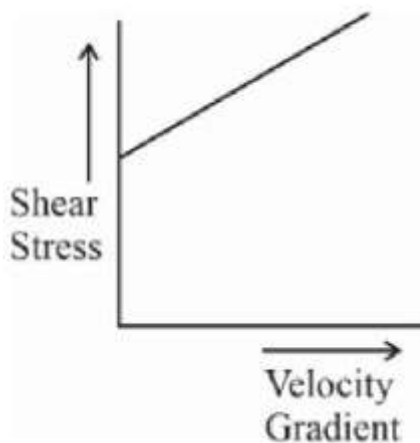


✓ 2. If one liter of a fluid has a mass of 7.5 kg then its specific gravity is 2/2

- ☐ 0.75
- ☒ 7.5
- ☐ 75
- ☐ 750



✓ 1. Which type of fluid is shown in the given graph 2/2



- ☐ Newtonian Fluid
- ☒ Bingham Plastic Fluid
- ☐ Ideal Fluid
- ☐ Real Fluid



Name *



✓ 7. What is the SI Unit of Surface Tension?

2/2

- ☒ N/m
- ☐ N/m²
- ☐ N/m³
- ☐ Unitless



✓ 11. Which one of the following law is applicable to a hydraulic lift?

2/2

- ☐ Kirchhoff's Law
- ☒ Pascal's Law
- ☐ Archimedes' Principle
- ☐ Archimedes' Law



✓ 14. At which of the following temperatures, the value of surface tension of water is minimum. 2/2

- ☐ 4° C
- ☐ 25 °C
- ☐ 50 °C
- ☒ 75 °C



✓ 5. The specific volume of a liquid is the reciprocal of

2/2

- ☐ weight density
- ☒ mass density
- ☐ specific weight
- ☐ specific volume



✓ 15. If diameter of a capillary tube increases then capillary rise will

2/2

- ☐ increase
- ☒ decrease
- ☐ remains same
- ☐ none of the above



✓ 10. If a tire gauge reads 35 psi, what is the absolute pressure inside the tire, assuming atmospheric pressure is 14.7 psi? 2/2

- ☐ 14.7 psi
- ☐ 20.3 psi
- ☐ 35 psi
- ☒ 49.7 psi



✓ 12. The Viscosity of Liquid decreases

2/2

- ☒ with increase of temperature
- ☐ with decrease of temperature
- ☐ may increase or decrease
- ☐ none of the above



✓ 3. Rain drops are spherical because of

2/2

- ☐ Viscosity
- ☐ Air Resistance
- ☒ Surface Tension Forces
- ☐ Atmospheric Pressure



✓ 8. Velocity distribution of fluid flow over a flat plate is given by equation $3 + 0.5y - y^2$ (where y is the vertical distance from the plate surface). Select the correct value of shear stress at a vertical distance of 0.2 m if the dynamic viscosity of the fluid is 0.7 N-s/m^2

2/2

- ☐ 0.14 N/m^2
- ☒ 0.07 N/m^2
- ☐ 0.1 N/m^2
- ☐ 0.28 N/m^2



✓ 6. In which of the following fluid the shear stress is directly proportional to the rate of shear strain? 2/2

- ☒ Newtonian
- ☐ Non-Newtonian
- ☐ Ideal
- ☐ Ideal Plastic



✓ 13. What is the SI unit of Viscosity?

2/2

- ☐ Poise
- ☒ Pascal-Second
- ☐ Newton-Second
- ☐ Newton/Second



Roll No. *

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✓ 9. Which law states that the rate of increase of pressure in a vertical direction is equal to weight density of fluid at that point? 2/2

- ☐ Pascal's Law
- ☐ Isothermal Law
- ☒ Hydrostatic Law
- ☐ Newton's law of Fluids



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